

Activity03 : Log Analysis

Part I. Can you find people trying to break into the servers?

Answer the following questions and provide evidence with your answer.

Q1. How many hackers are trying to get access to our servers? And how many attempts are there? Explain/define how you count distinct hackers.

Answer Number of distinct hackers = 185

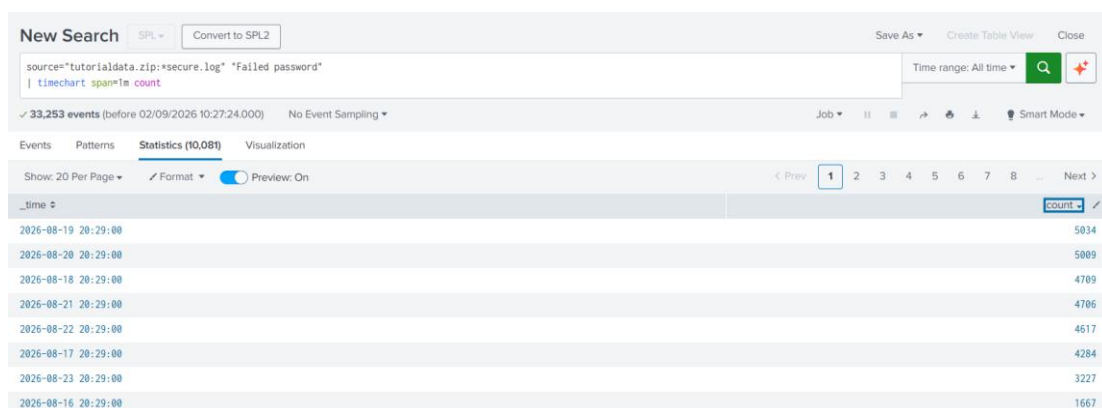
Number of attempts = 33253

Method to count distinct hackers show in image below.



Q2. What time do hackers appear to try to hack our servers?

20:29



Q3. Which server (mailsv, www1, www2, www3) had the most attempts?

From the image below it shows that **www1** had the most attempts.

New Search SPL Convert to SPL2 Save As Create Table View Close

source="tutorialdata.zip:*secure.log" "Failed password"
| stats count by source
| sort -source

Time range: All time Q +

✓ 33,253 events (before 02/09/2026 10:38:08.000) No Event Sampling Job || → ⬇ ⬆ Smart Mode

Events Patterns **Statistics (4)** Visualization

Show: 20 Per Page ✓ Format Preview: On

source	count
tutorialdata.zip:./www1/secure.log	8798
tutorialdata.zip:./www3/secure.log	8267
tutorialdata.zip:./mailsv/secure.log	8154
tutorialdata.zip:./www2/secure.log	8034

Q4. What is the most popular account that hackers use to try to break in?

root

New Search SPL Convert to SPL2 Save As Create Table View Close

source="tutorialdata.zip:*secure.log" "Failed password"
| rex field=_raw "Failed password for (invalid user)?(?<user>\S+) from"
| top user limit=10

Time range: All time Q +

✓ 33,253 events (before 02/09/2026 10:39:19.000) No Event Sampling Job || → ⬇ ⬆ Smart Mode

Events Patterns **Statistics (10)** Visualization

Show: 20 Per Page ✓ Format Preview: On

user	count	percent
root	1493	4.514802
administrator	1020	3.084460
admin	938	2.836493
operator	923	2.791134
mail	753	2.277057
mailman	752	2.274033
irc	644	1.947443
email	626	1.893012
games	601	1.817412
sys	586	1.772052

Part II. Sensitive Files on Web Servers

Q5. Can you find attempts to get access to sensitive information from our web servers? How many attempts were there?

Yes. There are attempts to access potentially sensitive information on the web servers, as indicated by unusual HTTP status codes such as **403 (Forbidden)**, **404 (Not Found)** in the access logs. Requests resulting in 403 suggest attempts to access restricted resources, while repeated 404 responses may indicate users or automated clients probing for files or directories that do not exist. The presence of these abnormal status codes, especially when associated with suspicious or sensitive-looking URLs, provides evidence of attempts to access potentially sensitive information.

New SearchSPLConvert to SPL2Save AsCreate Table ViewClose

```
source="tutorialdata.zip:access.log"
| rex field=_raw "\s*(?<method>\S+) (?<uri_path>[^\s\\?\""]+)"
| rex field=_raw "\s*" (?<status>\d{3}) "
| search status IN (403,404)
| stats count as attempts, values(status) as status_codes by uri_path
| sort - attempts
```

Time range: All time

✓ 918 events (before 02/09/2026 14:25:18.000) No Event SamplingJob

EventsPatternsStatistics (12)Visualization

Show: 20 Per PageFormatPreview: On

uri_path	attempts	status_codes
/product.screen	135	403 404
show.do	90	404
/stuff/logo.ico	84	404
/productscreen.html	82	404
/hidden/anna_nicole.html	73	404
/numa/numa.html	72	404
/rush/signals.zip	71	404
/search.do	70	404
/passwords.pdf	68	404
/category.screen	59	403
/oldlink	58	403
/cart.do	56	403

Q6. What resource/file are hackers looking for?

From Q5, you can see some sensitive files that hackers may be looking for. Such as

- /hidden/anna_nicole.html
- /numa/numa.html
- /rush/signals.zip
- /password.pdf

Part III. Are there bots crawling our websites?

Q7. Can you find any bots crawling our websites?

Yes, I query all user agent that including word “bot” and found this.



The screenshot shows a log analysis interface. At the top, a query is entered in a text box: `source=tutorialdata.zip:*access.log`, `| rex field=_raw "\(<method>\$+ \(<uri_path>[^\s\\?\\"]*\>)"`, `| rex field=_raw "\(<useragent>[^\"]*\>)"`, `| search useragent="*bot"`, `| stats count as requests by useragent`, `| sort - requests`. To the right of the query box is a search icon and a time range dropdown set to "All time". Below the query box, it says "1,849 events (before 02/09/2026 11:08:23.000) No Event Sampling". Below this is a navigation bar with tabs for "Events", "Patterns", "Statistics (4)", and "Visualization". Under "Statistics (4)", there are options for "Show: 20 Per Page", "Format", and "Preview: On". The main content area is a table with two columns: "useragent" and "requests". The table contains four rows of data.

useragent	requests
Mozilla/5.0 (compatible; Googlebot/2.1; +http://www.google.com/bot.html)	532
Googlebot/2.1 (http://www.googlebot.com/bot.html)	495
Googlebot/2.1 (http://www.googlebot.com/bot.html)	439
Mozilla/5.0 (compatible; YandexBot/3.0; +http://yandex.com/bots)	383

Q8. What are they doing on the site? (Hint: Look for User-Agent in the web

access.logs.)

The bots are crawling the website to collect information and build an index for search engines. Their User-Agent identifies them as automated crawlers, which systematically visit and retrieve web pages so that the content can be indexed and made searchable by users.

New Search SPL Convert to SPL2 Save As Create Table View Close

Time range: All time Q +

```
source="tutorialdata.zip:access.log"
| rex field=_raw "(?<clientip>\$+)"
| rex field=_raw "\"(?<method>\$+) (?<url_path>[^\s?\\"]+)"
| rex field=_raw "\"(?<useragent>[^\"]+)\$"
| search useragent="*bot*"
| stats count as requests, values(uri_path) as pages by useragent
| sort - requests
```

✓ 1,849 events (before 02/09/2026 11:25:33.000) No Event Sampling Job || ↗ ⌵ Smart Mode

Events Patterns **Statistics (4)** Visualization

Show: 20 Per Page ✓ Format Preview: On

useragent	requests	pages
Mozilla/5.0 (compatible; Googlebot/2.1; +http://www.google.com/bot.html)	532	/cart.do /cart/error.do /cart/success.do /category.screen /hidden/anna_nicole.html /numa/numa.html /oldlink /product.screen /rush/signals.zip /stuff/logo.ico show.do
Googlebot/2.1 (http://www.googlebot.com/bot.html)	495	/cart.do /cart/error.do /cart/success.do /category.screen /hidden/anna_nicole.html /oldlink /product.screen